Kabeen Kim

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Research Interests: *Efficient LLM Inference · Resource-Constrained AI Systems · Faithful Retrieval and Generation*

Education

Duksung Women's University — *Seoul, South Korea*
B.S. in Computer Engineering & Statistics | GPA: 4.15 / 4.5 — *Mar 2021 – Feb 2027 (Expected)*

University of Mississippi — *Mississippi, USA*
Exchange Student, Computer Science — *Aug 2024 – Dec 2024*

Publications

[1] Haopeng Chen, Yihao Ai, **Kabeen Kim**, et al. "UDAPose: Unsupervised Domain Adaptation for Low-Light Human Pose Estimation." *Computer Vision and Pattern Recognition (CVPR), 2026.* [Paper] [Code]

[2] **Kabeen Kim**[*†], Yena Kim[*], Minjeong Seo[*]. "ConFuse: Context-Aware Fusion of LLMs and GBDTs for Sleep Prediction from Lifelog Data." *International Conference on ICT Convergence (ICTC), 2026.* (Poster) [Code] *[*]Equal contribution. [†]Presenter.*

[3] **Kabeen Kim**[†], et al. "CoDU: Multi-Stage Approach for Complex Document Structure Understanding." *Korean DataBase Conference (KDBC), 2025.* (Oral) [Code]

[4] **Kabeen Kim**[†], et al. "SPTC(Single Pass Tree Chain-of-thought): An Efficient Prompting Scheme for Small Language Models." *Korea Computer Congress (KCC), 2025.* (Poster) [Paper]

Research Experience

Electronics and Telecommunications Research Institute (ETRI) — *Daejeon, South Korea*
Research Intern, Government-Funded Research Institute — *Jul 2026 – Aug 2026*

Multi-hop GraphRAG – *Efficient Graph Construction and Evidence Retrieval*

- Built a ∼**38K-relation** knowledge graph from a technical manual and **250 multi-hop questions** to trace evidence loss.
- Improved graph construction with length-sorted batches, OOM batch-size backoff, and checkpointing; reduced runtime **from ∼3 to 1.5 days**, enabling evaluation across additional manuals.
- Redesigned path-based traversal and reranking: all required evidence retrieved for **202/250 questions, up from 62**.
- Reached **Recall@5 of 0.873** on 300 HotpotQA questions, 2nd among compared methods behind HippoRAG (0.898).
- Evaluated grounded answers with an LLM judge: **correctness 2.84 → 3.86/5**; system adopted in a lab project.

Data Mining & Knowledge Discovery Lab, Duksung Women's University — *Seoul, South Korea*
Undergraduate Researcher — *Mar 2025 – Feb 2026*

SPTC [4] – *Efficient Prompting for Small Language Models*

- Integrated generation, self-evaluation, and selection in one prompt; tested five small LLMs on two QA datasets.
- Ran **1.5× faster than ToT** at 1.4 pp lower accuracy and **21.3 pp above CoT**; traced errors to self-evaluation.

CoDU [3] – *2025 Samsung AI Challenge, 3rd of 210 Teams*

- Recruited and led a three-person team, designing the layout detection, OCR, and reading-order pipeline.
- Parallelized detection and OCR for the **fastest inference** in the competition.
- Labeled 322 documents to fine-tune DocLayout-YOLO (**F1 0.463 → 0.580**); layout improvements also raised downstream OCR and reading-order scores without modifying those modules.

Product Search RAG – *Lab Collaboration*

- Defined the evaluation, then tuned retrieval and reranking against it: **F1 0.3 → 0.6**, leading to a follow-up contract.

Vision and Multimodal Intelligence Lab, University of Mississippi — *Mississippi, USA*
Undergraduate Researcher — *Sep 2024 – Aug 2025*

UDAPose [1] – *Unsupervised Domain Adaptation for Low-Light Pose Estimation*

- Classified failures on **491 low-light images into 15 categories**; evaluated CycleGAN and UNSB image translation.
- Contributed to UDAPose [1]; the team's method beat the strongest baseline by **10.1 AP on ExLPose's hard subset**.

Work Experience

Boston Consulting Group (BCG) *Seoul, South Korea*

Research Analyst Intern *Mar 2026 – May 2026*

Domain LLM & RAG QA – *Fine-Tuning, Evaluation, and Slide Parsing*

- LoRA-tuned Qwen models for domain QA, raising **LLM judge scores from 2.3 to 3.3/5**; gains only in the largest model.
- Automated QA checks (deduplication, grounding) and built an evaluation framework with EM, token F1, and LLM judge.
- Built a slide parser and value-extraction agent; with Claude Sonnet 4.6, it **ranked 1st among tested parser–LLM pairs**.

Selected Projects

Dimension of Memory: Multimodel AI Serving on an 8GB GPU *github.com/4CO4CO/4co4co-ai*

AI Lead, 4-Person Capstone Team – Hanium Dream-Up (Government Mentoring Program) *Feb 2025 – Nov 2025*

- Integrated **five AI models on one 8GB GPU** into a photo-to-music pipeline for a three-day exhibition.
- Tested CPU/GPU placement; after Moondream2 failed on CPU, used GPU execution with one worker to avoid OOM.
- Measured **95.1% peak VRAM usage** and ∼**55 seconds/image** in the final, serialized configuration.

ConFuse: Context-Aware Fusion of LLMs and GBDTs *github.com/kbannie/ConFuse*

Research Team Lead, ToBig's Big Data & AI Society *May 2026 – Sep 2026*

- Led a team fusing GBDTs and LLMs for sleep prediction from small lifelog data, fine-tuning **Qwen2.5-7B with QLoRA**.
- Weighted the LLM per sample by input quality, reaching **0.6415 log loss**, below ML-only and 50:50 fusion [2].

Honors & Awards

2025 **3rd Place (out of 210 teams)**, Samsung AI Challenge – Visually-Rich Document Understanding
2025 **Encouragement Award**, DS Regional Collaboration Idea Competition
2024 **31st Mirae Asset Overseas Exchange Scholarship**, Mirae Asset Park Hyeon Joo Foundation
2024 **Excellence Award**, TAVE 13th Data Analysis Project
2024 **2nd Prize**, DS Global Vision Makers Innovation Idea Competition
2021–22 **Academic Excellence Scholarship (3 awards)**, Duksung Women's University

Leadership

Duksung Global Challenger

Team Lead & Project Proposer, "Be a Connector" (WeCon) *Mar 2025 – Sep 2025*

- Secured **KRW 10 million in funding** to build a BLE beacon app linking the campus with local businesses.
- Interviewed experts in 4 European countries in English on BLE/UWB tradeoffs, shaping app features and budget.

Teaching & Volunteering

Chungnam National University & NAQS (Korean Government Agency)

Teaching Assistant, n8n & RAG Workshop *Sep 2025 – Oct 2025*

- Authored **170 slides** and live demos on API authentication, Docker, and n8n for 20-hour lectures (50 participants).
- Prototyped a RAG Q&A lab and **3 workflow automation labs** with n8n for public-sector staff.

CJ OliveNetworks SW Creative Camp

University Volunteer Instructor, CJ UNIT (8th Cohort) *Mar 2023 – Jun 2023*

- Taught **40 hours of LEGO robotics and Python coding** to middle school students in Jeju via remote classes.
- Co-designed a 3-session ESG curriculum on wave, solar, and wind energy with LEGO guides and Python examples.

Skills

- **Programming & Statistics**: Python, R, C++, SQL (SQLD certified); regression, time series, and multivariate analysis
- **ML & Deep Learning**: PyTorch, Unsloth (LoRA/QLoRA), OpenCV, YOLO, XGBoost, LightGBM
- **LLM & Retrieval**: LangChain, Qdrant, Pinecone, NetworkX
- **Infrastructure & Tools**: PostgreSQL, AWS (EC2/S3), Docker, Git, Weights & Biases, n8n